

AI Team Requirements (Python Microservice)

Objective: Build a FastAPI service that acts as the "Brain" for the SigmaLoop platform. It handles two distinct tasks:

1. **Lesson Hints:** Using RAG to answer specific questions about lesson text.
2. **Global Mentor:** Using "Context Injection" to act as a personalized tutor.

Tech Stack

- **Language:** Python 3.10+
- **Framework:** FastAPI
- **Database:** ChromaDB (Local vector store)
- **LLM:** Groq (Llama 3) or OpenAI (GPT-3.5/4)

Task 1: The Ingestion Pipeline (RAG Setup)

Goal: Prepare the lesson data so the AI can "read" it.

1. Create a folder data/ and add lesson1.txt (Variables), lesson2.txt (Loops).
2. Write a script ingest.py that:
 - Loads these text files.
 - Splits them into chunks (Chunk size: 500 chars).
 - Embeds them using HuggingFaceEmbeddings.
 - Saves them to ./chroma_db.
 - **Metadata:** Ensure each chunk has metadata { "lesson_id": "L1" } so we can filter later.

Task 2: Endpoint A - Lesson Hints (RAG)

Use Case: User is on the "Single Lesson Page" and clicks the floating "Ask AI" button.

Endpoint: POST /generate-hint

Logic:

1. Receive question and lesson_id.
2. Query ChromaDB for text chunks matching question, strictly filtering where metadata.lesson_id == lesson_id.
3. Combine chunks + question into a prompt.
4. Return the answer.

Task 3: Endpoint B - Global Mentor (Context Aware)

Use Case: User is on the "Mentor Chat Page" and asks for help.

Endpoint: POST /mentor/chat

Request Body (This is what Node.js sends you):

```
{
  "user_message": "I don't understand why I failed.",
  "chat_history": [
    {"role": "user", "content": "Hi"},
    {"role": "assistant", "content": "Hello!"}
  ],
  "context": {
    "user_name": "Ahmed",
    "current_lesson": "Python Loops",
    "last_error": "IndentationError: expected an indented block",
    "last_code_snippet": "for i in range(5):\nprint(i)"
  }
}
```

Implementation Logic:

You do NOT need RAG for this specific task (unless they ask a specific curriculum question).

You rely on the Context provided in the JSON.

The System Prompt:

You are SigmaBot, a friendly coding mentor.

You are talking to {context.user_name}.

CURRENT STATUS:

The user is working on the lesson: {context.current_lesson}.

Their last run resulted in this error: {context.last_error}.

Here is their code:

{context.last_code_snippet}

USER MESSAGE:

{user_message}

INSTRUCTIONS:

1. If the user asks about their error, explain it using the code snippet provided.
2. Be encouraging.
3. Keep answers concise.

Task 4: Definition of Done

1. **FastAPI Server** is running.

2. **RAG works:** Sending a question about "Variables" returns an answer from lesson1.txt.
3. **Mentor works:** Sending a JSON with a fake "IndentationError" results in the AI explaining how to fix indentation.